

M5-TTS: A Mini-Scale Multilingual and Multi-Dialect Text-to-Speech Language Model with Multi-Token Prediction

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https://github.com/ASLP-lab/M5-TTS
https://huggingface.co/ASLP-lab/M5-TTS
https://aslp-lab.github.io/M5-TTS-Demo-Page
https://huggingface.co/ASLP-lab/M5-TTS-Space

Abstract

We present **M5-TTS**, a compact text-to-speech system supporting multilingual and Chinese multidialect synthesis, zero-shot voice cloning, speaker-specific multilingual generation, and CPU streaming. Its **50M-parameter language model**, excluding the speech codec, combines a temporal backbone with a local multi-token prediction module over 16 codebooks. GRPO post-training combines global speaker-similarity and pause feedback with a token-level CER process advantage. Character-level error comparisons provide local feedback, with negative credit propagated backward within a clause and projected onto audio-token positions. M5-TTS achieves 1.37% CER on Seed-ZH and 2.91% WER on Seed-EN, corresponding to relative error reductions of 62.2% and 35.3% over MOSS-TTS-Nano with half the language-model parameters. It also achieves lower content error rates in five of six evaluated MiniMax languages, while difficult Chinese inputs remain a limitation. On four Intel Xeon Gold 6530 cores, it achieves a real-time factor of 0.653 and a warm first-chunk latency of 225 ms. A 10-listener evaluation gives M5-TTS higher speaker-similarity and intelligibility ratings than MOSS-TTS-Nano, with a lower naturalness rating. We will release training, post-training, and inference code, together with voice-cloning and speaker-specific checkpoints, to support reproducible research on compact speech generation for on-device voice interaction.

1 Introduction

Speech generation is a key component of spoken interaction for embodied agents, personal assistants, and intelligent devices. In these settings, a speech synthesizer needs to produce intelligible and natural speech, maintain a consistent voice, and begin playback with a short delay. Multilingual and dialectal capabilities further allow a single device to communicate with users across linguistic communities. Local inference can reduce dependence on network connectivity and give developers greater control over the speech generation pipeline. These requirements motivate compact text-to-speech (TTS) systems that

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balance synthesis quality, linguistic coverage, voice flexibility, and streaming efficiency under limited computational resources.

Neural TTS research has explored diverse generation architectures. Early end-to-end approaches, such as VITS Kim et al. (2021), combine variational inference and adversarial learning for waveform generation, while non-autoregressive models such as FastSpeech 2 Ren et al. (2021) improve synthesis efficiency through parallel acoustic modeling and explicit duration, pitch, and energy conditioning. However, point-wise acoustic regression objectives used in some non-autoregressive systems can encourage over-smoothing Ren et al. (2022), reducing acoustic and prosodic variation. Richer generative models aim to capture this variation more effectively.

Language-model-based TTS approaches use speech tokens as intermediate representations Wang et al. (2023), enabling autoregressive speech generation through sequence modeling. Subsequent studies have explored diverse speech representations and generation paradigms. CosyVoice 2 Du et al. (2024) combines autoregressive language modeling with conditional flow matching, while Qwen3-TTS Hu et al. (2026) employs multi-codebook tokens for multilingual and streaming synthesis. Recent work has also explored continuous speech generation without relying on conventional discrete tokens. DiTAR Jia et al. (2025) combines an autoregressive language model that captures dependencies across speech patches with a local diffusion Transformer that generates continuous acoustic features within each patch. VoxCPM2 Zhou et al. (2026) further explores tokenizer-free diffusion autoregressive modeling for multilingual speech generation and controllable voice cloning. Despite these advances, the substantial model sizes and inference costs of these systems can make low-latency streaming challenging on resource-constrained edge devices, particularly under CPU-only inference.

The growing demand for on-device voice interaction and embodied intelligence therefore creates a pressing need for compact TTS models that preserve naturalness and expressiveness while supporting efficient, responsive streaming synthesis. Recent compact TTS systems have further advanced on-device speech generation. MOSS-TTS-Nano OpenMOSS Team (2026) explores lightweight language-model-based speech generation with discrete codec tokens, while Pocket TTS Rouard et al. (2026) demonstrates efficient CPU-based inference with compact continuous models. Other lightweight projects, including Audio8-TTS Preview Audio8 Team (2026), Kokoro-TTS hexgrad (2025), Hojo-TTS-Light HojoAI (2026), TinyTTS Lê (2026), and sanoTTS Thapa (2026), further investigate reducing model complexity and computational requirements for resource-constrained speech synthesis.

Combining broad language and dialect coverage, zero-shot voice cloning, and CPU streaming within a 50M-parameter language model remains a challenging design objective. We study this combination under a shared, compact generation architecture.

We present **M5-TTS**, a mini-scale multilingual and multidialect TTS language model with multi-token prediction (MTP), built around 50M parameters excluding the speech codec. A temporal backbone and a local Transformer jointly generate a 16-codebook speech representation. Language and accent conditioning support linguistic control, while reference speech enables zero-shot cloning and speaker-specific fine-tuning provides a dedicated multilingual voice. To refine synthesis, we use Group Relative Policy Optimization (GRPO) Shao et al. (2024) with two complementary feedback paths. Global SIM and pause rewards assess overall speaker resemblance and phrasing. A local CER process advantage compares character errors across candidate utterances, propagates negative feedback backward within a clause, and maps it to audio-token positions. Adding this local term to the global advantage provides content feedback at the generation decisions associated with each text region.

Our experiments examine synthesis quality and streaming performance, with MOSS-TTS-Nano OpenMOSS Team (2026) as the main comparison among lightweight language-model-based TTS systems. Using half as many language-model parameters, M5-TTS achieves 1.37% CER on Seed-ZH and 2.91% WER on Seed-EN, compared with 3.62% and 4.50% for MOSS-TTS-Nano, respectively. On four cores of an Intel Xeon Gold 6530, M5-TTS achieves a real-time factor of 0.653 and a warm first-chunk latency of 225 ms. In the 10-listener study, M5-TTS receives higher speaker-similarity and intelligibility scores than MOSS-TTS-Nano, although its naturalness score is lower. We will release all training, post-training, and inference code, together with zero-shot voice-cloning and speaker-specific checkpoints, to support reproducible research and further community development. With this work, we aim to provide a practical speech generation component for on-device applications and embodied voice interaction.

2 Architecture

2.1 Tokenizer

We adopt Qwen-TTS-Tokenizer-12Hz from Qwen3-TTS Hu et al. (2026) to convert speech into discrete representations at 12.5 frames per second. Each frame contains 16 codebook indices, comprising one

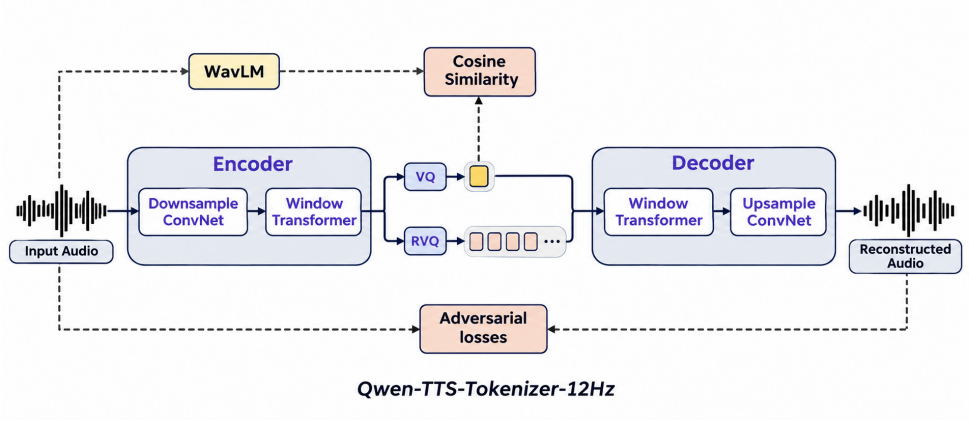


Figure 1: Qwen-TTS-Tokenizer-12Hz Hu et al. (2026) adopted by M5-TTS: a semantic VQ branch and an acoustic RVQ branch provide 16 codec tokens per frame for causal waveform reconstruction.

semantic codebook and 15 acoustic codebooks. The encoder consists of a downsampling convolutional network followed by a windowed Transformer. Its latent features are quantized through a semantic VQ branch and a 15-stage acoustic RVQ branch, which represents complementary acoustic information. A windowed Transformer and an upsampling convolutional network then reconstruct the waveform from the quantized representations. In the tokenizer's original training framework, WavLM Chen et al. (2021) provides teacher features for semantic distillation of the first codebook using a cosine-similarity objective. Adversarial training encourages perceptually faithful reconstruction, while a multi-scale mel-spectrogram loss constrains spectral consistency. Both the encoder and decoder operate causally: speech tokens are extracted without look-ahead, and the corresponding waveform is reconstructed incrementally. This combination of a low frame rate and causal reconstruction provides M5-TTS with a compact speech representation suited to streaming generation.

2.2 Backbone

Following the hierarchical Transformer organization of Marvis-TTS Canuma & Newman (2025) and the multi-codebook prediction scheme of Qwen3-TTS Hu et al. (2026), M5-TTS uses a compact language-modeling architecture with 50M parameters, excluding the speech codec. The architecture comprises a causal backbone Transformer for temporal modeling and a lightweight local Transformer serving as the MTP module. The conditioning interface orders accent, speaker, and language IDs before the text sequence; the speaker ID is used in speaker-specific fine-tuning, whereas pre-training uses accent and language IDs. For each historical audio frame, codebook-specific embeddings of its 16 codec tokens are summed into a single frame-level input, allowing the backbone to model speech at the codec frame rate.

Generation follows a hierarchical codebook prediction scheme. At each time step, the backbone summarizes the conditioning sequence and previously generated audio frames into a hidden state. A dedicated output head first predicts the zeroth codebook token. The backbone state and this token's embedding are projected to the MTP input space. The MTP module then autoregressively generates codebooks 1–15, with each prediction conditioned on the preceding codebooks within the same frame. The completed frame is aggregated and fed back to the backbone for the next temporal step.

During inference, the backbone retains its attention cache across audio frames, while the MTP cache is reset for each new frame. This factorization separates temporal modeling from within-frame acoustic refinement and avoids expanding all 16 codebooks into separate backbone time steps. Completed codec frames are passed to the causal waveform decoder for incremental reconstruction, supporting streaming synthesis.

3 Methods

3.1 Pre-training

3.1.1 Basic Pre-training

M5-TTS is pre-trained on approximately 400,000 hours of speech. The corpus combines the multilingual Emilia and Emilia-YODAS collections He et al. (2024; 2025), approximately 160,000 hours of in-house recordings, and WenetSpeech4TTS, WenetSpeech-Yue, and WenetSpeech-Chuan Ma et al. (2024); Li et al.

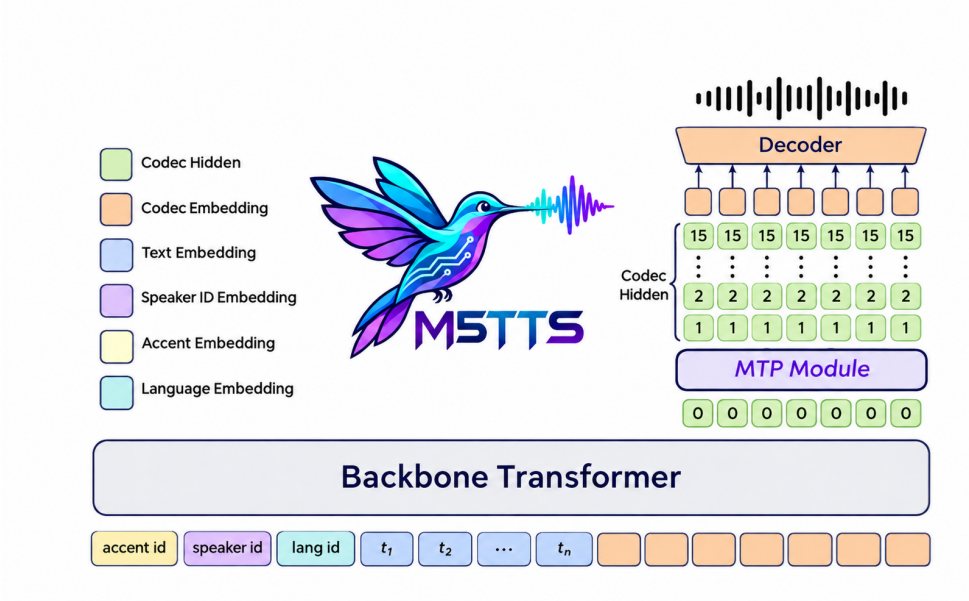


Figure 2: M5-TTS architecture. The temporal backbone predicts codebook 0; the local MTP module generates codebooks 1–15 within each frame, which are then decoded into speech.

(2026); Dai et al. (2025). Together, these sources cover Chinese, English, German, French, Japanese, and Korean, as well as regional Chinese speech.

Pre-training conditions on the accent ID, language ID, and target text, in that order. Speaker IDs are introduced during speaker-specific fine-tuning. Speech is represented by 16 codebook tokens per codec frame. Supervised training learns text–speech correspondences under the hierarchical autoregressive factorization: the temporal backbone predicts the zeroth codebook, and the local MTP module predicts the remaining codebooks conditioned on the backbone state and preceding within-frame tokens. Teacher forcing supplies the observed speech history during likelihood training. This establishes a general speech-generation policy for subsequent refinement of pronunciation, speaker consistency, and speaker-specific synthesis.

3.1.2 Continued Pre-training

To improve acoustic quality, expressiveness, and pronunciation robustness in multilingual and multidialectal synthesis, we perform continued pre-training (CPT) on a curated subset of the pre-training corpus. This subset comprises approximately 65,000 hours of speech, with selection prioritizing high-quality recordings, expressive speech, and challenging samples involving complex pronunciation patterns. The corpus covers Chinese, English, German, French, Japanese, and Korean, together with dialect data including Cantonese and Sichuanese. Language and dialect tags are encoded alongside the target text to provide explicit conditioning for speech generation.

3.2 Post-training

3.2.1 Group Relative Policy Optimization

We use Group Relative Policy Optimization (GRPO) Shao et al. (2024) to combine overall synthesis feedback with local feedback on content errors. For the same target text and reference speech, the rollout policy samples G candidate utterances. Each candidate receives an utterance-level score for speaker similarity and pausing, together with character-level feedback derived from ASR errors. The global advantage is shared across the utterance; the local CER advantage is added at the corresponding audio-token positions. Figure 3 summarizes this two-branch procedure.

Global feedback. Speaker similarity measures resemblance to the reference voice. The pause reward measures whether aligned inter-character gaps satisfy the duration constraints for ordinary and punctuation-associated boundaries. We obtain timestamps with Qwen3-ForcedAligner Shi et al. (2026); the pause-score definition is given in Appendix A.3. We combine the two scores and normalize the

resulting reward within each candidate group:

$$\begin{aligned} R_i^{\text{global}} &= \lambda_{\text{SIM}}\,\text{SIM}_i + \lambda_{\text{pause}} R_i^{\text{pause}}, \\ A_i^{\text{global}} &= \frac{R_i^{\text{global}} - \mu_R}{\sigma_R + \varepsilon}, \end{aligned} \tag{1}$$

where μ_R and σ_R are the group mean and standard deviation, and ε ensures numerical stability. We use $\lambda_{\text{SIM}} = 1.0$ and $\lambda_{\text{pause}} = 0.5$. Sentence-level CER is not broadcast as an additional global reward; content feedback enters through the local branch below.

Character-level process feedback. We align each candidate's ASR transcript with the target text using edit alignment. Let $e_{i,j}$ be the number of edit operations assigned to target character j in candidate i. Substitutions and deletions are assigned to the affected character; insertions are assigned to the following target character, or to the last character for trailing insertions. At each target position, we compare the edit counts across the G candidates:

$$a_{i,j} = \frac{\mu_j - e_{i,j}}{\sigma_j + \varepsilon}, \tag{2}$$

where μ_j and σ_j are the group mean and standard deviation of the edit counts at position j. Fewer errors than the group mean produce positive local feedback; more errors produce negative feedback. If all candidates have the same edit count at a position, its unpropagated local advantage is zero.

An error may depend on generation decisions made before its audible location. We therefore propagate negative character advantages backward within the same clause, with exponential decay over character distance:

$$\widetilde{a}_{i,j} = a_{i,j} + \sum_{k \in \mathcal{C}_j^+} \exp\big(-(k-j)\big)\,\min(a_{i,k}, 0), \tag{3}$$

where $\mathcal{C}_j^+$ contains the characters after j in its clause. Only negative advantages are propagated. Earlier characters can thus receive attenuated feedback from a later error, even when they were recognized correctly.

From characters to audio tokens. Forced alignment provides timestamps for the recognized text, and edit alignment associates these intervals with target characters. We project the propagated advantages onto codec frames. For a character with an acoustic interval, its projection weight $w_{i,t,j}$ is the overlap duration with frame t, divided by the frame duration. If the weights within a frame sum to more than one, they are normalized by that sum. A frame entirely inside one character interval therefore receives the full character advantage, rather than a share divided by the number of frames in that interval. Deletions have no spoken interval, so their feedback is attached to an available following boundary decision or the terminal decision. The projection weights also encode these deletion-to-boundary assignments.

The resulting local process advantage is added to the global advantage:

$$\begin{aligned} A_{i,t}^{\text{local}} &= \sum_j w_{i,t,j}\,\widetilde{a}_{i,j}, \\ A_{i,t} &= A_i^{\text{global}} + \lambda_{\text{CER}} A_{i,t}^{\text{local}}, \qquad \lambda_{\text{CER}} = 0.5. \end{aligned} \tag{4}$$

All codebooks in a frame share $A_{i,t}$. Thus, every generated decision receives the global synthesis feedback, while the local branch adds position-dependent content feedback.

Policy optimization. For codebook token $o_{i,t,c}$ with conditioning context $h_{i,t,c}$, the importance ratio compares the current policy with the rollout policy that generated the candidate:

$$r_{i,t,c}(\theta) = \frac{\pi_\theta(o_{i,t,c} \mid h_{i,t,c})}{\pi_{\text{old}}(o_{i,t,c} \mid h_{i,t,c})}. \tag{5}$$

We minimize a clipped policy loss with KL regularization:

$$\begin{aligned} \ell_{i,t,c}(\theta) &= -\min\{r_{i,t,c}(\theta)A_{i,t},\ \text{clip}(r_{i,t,c}(\theta), 1-\epsilon_-, 1+\epsilon_+)\,A_{i,t}\}, \\ \mathcal{L}_{\text{GRPO}}(\theta) &= \mathcal{L}_{\text{clip}}(\theta) + \beta \mathcal{L}_{\text{KL}}(\theta), \qquad \beta = 0.01. \end{aligned} \tag{6}$$

Here ϵ_- and ϵ_+ are the clipping margins, and $\mathcal{L}_{\text{clip}}$ aggregates the token losses $\ell_{i,t,c}$. Ratios and clipping are computed separately for each codebook. The KL term constrains deviation from a frozen supervised reference policy π_{ref}, which is distinct from π_{old}. Advantages and reference predictions are detached during optimization. Loss aggregation and the rollout configuration are described in Appendices A.3 and A.4.

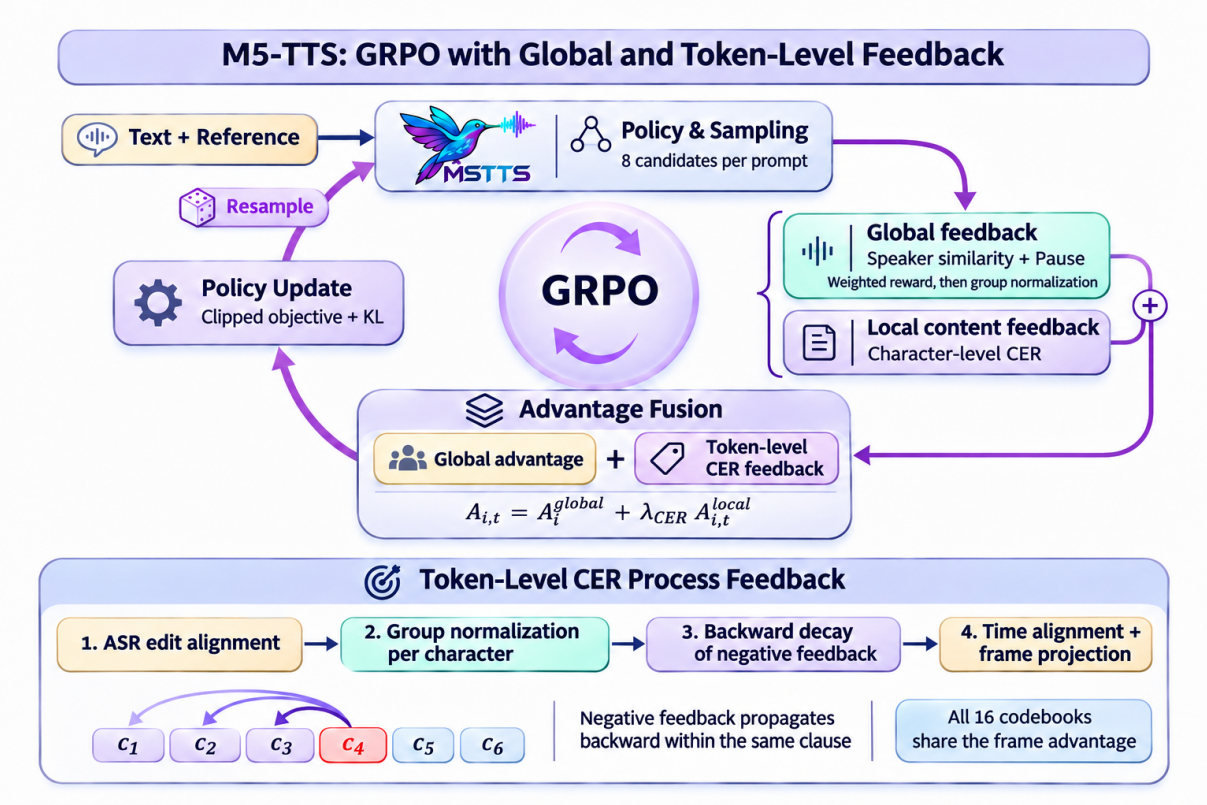


Figure 3: GRPO with global and token-level feedback. Utterance-level speaker-similarity and pause rewards yield a shared global advantage. ASR edit alignment yields per-character group-relative advantages; negative advantages propagate backward within a clause before temporal projection onto codec frames. The local CER process advantage is added to the global advantage for the policy update.

3.2.2 Speaker-Specific Fine-Tuning

Speaker-specific fine-tuning adapts the shared model to curated recordings of a designated speaker. A speaker identifier is added between the accent and language identifiers, retaining the hierarchical token-prediction objective. Curated recordings from the M2VoC Challenge 2021 Xie et al. (2021) provide speaker-specific supervision.

This setting evaluates an adapted voice rather than matching a supplied reference speaker at inference time. We therefore report custom-voice and zero-shot voice-cloning results separately.

3.3 Efficiency Optimization

To enable low-latency streaming speech synthesis on both CPUs and GPUs, we optimize the inference pipeline of M5-TTS without modifying its hierarchical autoregressive architecture. Our optimizations focus on four aspects: computation graph optimization, stateful inference and cache reuse, adaptive streaming scheduling, and hardware-aware runtime optimization.

3.3.1 Computation Graph Optimization

We reduce inference overhead by restructuring the execution graphs of the backbone and depth decoder. After the backbone predicts the first codebook, we encapsulate prediction and sampling of the remaining 15 codebooks within a single local-decoder frame graph, reducing Python-level iteration and runtime invocation overhead. Autoregressive dependencies among these codebooks are preserved.

We further fuse the query, key, and value projections in attention layers, as well as the gate and up projections in feed-forward layers, to reduce redundant operations and intermediate memory accesses. In addition, we precompute the linear projections of audio embeddings, replacing repeated embedding projection operations with efficient lookup-table accesses.

3.3.2 Stateful Inference and Cache Reuse

To eliminate redundant computation during autoregressive generation, we introduce explicit KV-cache management for both the backbone and depth decoder. On CPUs, these states are explicitly passed between ONNX Runtime inference calls, enabling incremental decoding and independent ONNX deployment.

We also implement a two-level reference caching mechanism. Reference audio is first encoded into reusable codec tokens, while the CPU inference pipeline additionally caches decoder states associated with complete reference inputs. A bounded LRU policy controls cache memory usage.

For streaming waveform generation, we implement a stateful codec decoder that maintains causal convolution histories, transposed-convolution overlap states, and sliding-window attention KV caches. This allows subsequent decoding steps to process newly generated codec tokens while reusing historical states, reducing redundant computation across overlapping audio chunks.

3.3.3 Adaptive Streaming Scheduling

We adopt a latency-aware streaming strategy that separates initial-chunk generation from subsequent decoding. A smaller initial chunk is used to reduce time to first audio, while larger decoding batches improve throughput during continuous generation.

For CPU inference, the decoding batch size is dynamically selected from 4, 8, and 16 frames according to the amount of generated audio available ahead of playback. This adaptive scheduling strategy balances first-chunk latency, overall real-time factor, and playback continuity.

3.3.4 Hardware-Aware Runtime Optimization

We apply hardware-specific execution optimizations to improve inference efficiency across CPU and GPU platforms.

For CPU inference, we optimize ONNX Runtime execution through physical-core affinity, intra-operation thread tuning, single-threaded inter-operation scheduling, sequential execution, and graph optimization. We additionally reduce resource overhead through idle thread-pool suspension, memory-mapped embeddings, and preallocated generation buffers. The execution graph is selected according to the CPU and threading configuration.

For GPU inference, the TTS model operates in BF16 precision, while the codec decoder remains in FP32 to preserve numerical stability. We compile the depth-frame computation graph and codec decoding operations using `torch.compile`, with separate compiled codec graphs for fixed input-window shapes.

Finally, we reduce CPU–GPU synchronization and memory allocation overhead by accumulating EOS indicators within each decoding chunk and checking them at chunk boundaries, while preallocating codec-token, single-frame input, and positional buffers. The fast inference configuration additionally employs scaled dot-product attention (SDPA) for efficient attention computation.

4 Experiments

4.1 Benchmarks and Metrics

We evaluate on Seed-TTS Eval ByteDance Speech (2024), the MiniMax TTS-Multilingual-Test-Set Zhang et al. (2025), and the Cantonese and Sichuanese benchmarks from WenetSpeech-Yue and WenetSpeech-Chuan Li et al. (2026); Dai et al. (2025).

Content accuracy is measured by character error rate (CER) or word error rate (WER), using the edit distance between an ASR transcription of the generated speech and the target text. For either unit of comparison, the error rate is $100(S + D + I)/N$, where S, D, and I denote substitutions, deletions, and insertions, and N is the number of reference units. We use CER for Chinese, Japanese, and Korean and WER for English, German, and French. Speaker similarity (SIM) measures consistency with the reference voice, and higher values are preferred.

Systems and settings. We distinguish Voice Cloning and Custom Voice configurations throughout. The latter uses a specific speaker configuration and is not a matched zero-shot cloning comparison. Additional lightweight baselines are included where measurements are available. Missing entries indicate unavailable or unreported measurements; they do not establish that a capability is unsupported. Bold model names identify M5-TTS.

4.2 Objective Results

Table 1: Structure of the evaluation benchmarks. CER and WER are reported in percent.

Benchmark	Subset	Language(s)	Content metric ↓	Speaker metric ↑
Seed	Seed-ZH	ZH	CER (%)	SIM
	Seed-EN	EN	WER (%)	SIM
	Seed-Hard	ZH	CER (%)	SIM
MiniMax	Multilingual	ZH, JA, KO	CER (%)	SIM
	Multilingual	EN, DE, FR	WER (%)	SIM
WenetSpeech	Chuan	Sichuanese	CER (%)	SIM
	Yue	Cantonese	CER (%)	SIM

Notes. ZH: Chinese; EN: English; DE: German; FR: French; JA: Japanese; KO: Korean.

Table 2: Results on Seed-TTS Eval. CER and WER are reported in percent. Lower CER/WER and higher SIM indicate better performance.

Model	Main Model Parameters	Setting	Seed-ZH		Seed-EN		Seed-Hard	
			CER↓	SIM↑	WER↓	SIM↑	CER↓	SIM↑
M5-TTS (CPT)	50M	Voice Cloning	1.70	0.64	3.60	0.53	20.92	0.57
M5-TTS (GRPO)	50M	Voice Cloning	1.37	0.65	2.91	0.53	17.48	0.59
M5-TTS	50M	Custom Voice	1.26	—	1.88	—	18.19	—
Audio8-TTS Preview	170M	Voice Cloning	**1.21**	**0.68**	1.96	**0.57**	19.63	**0.63**
MOSS-TTS-Nano	100M	Voice Cloning	3.62	0.64	4.50	0.50	15.77	**0.63**
Pocket TTS	100M	Voice Cloning	—	—	**1.86**	0.47	—	—
Hojo-TTS-Light	80M	Voice Cloning	2.15	0.40	3.63	0.17	21.10	0.36
Pocket TTS	100M	Custom Voice[a]	—	—	1.90	—	—	—
Kokoro-TTS	82M	Custom Voice[b]	1.40	—	2.04	—	**6.33**	—
Hojo-TTS-Light	40M	Custom Voice[c]	1.78	—	2.97	—	22.70	—
Sano TTS	2.27M	Custom Voice[d]	2.42	—	2.40	—	7.08	—
Tiny TTS	1.6M	Custom Voice	—	—	3.30	—	—	—

Notes. — denotes an unavailable result or an unspecified setting. Setting distinguishes Voice-Cloning and Custom-Voice configurations. For models with multiple preset voices, Custom-Voice results are obtained using one selected voice per language, as specified below. Unmarked Custom-Voice models use a single fixed voice. The Audio8-TTS row uses the Preview-0.1B checkpoint Audio8 Team (2026); parameter counts refer to main models and exclude speech codecs.

[a] The English voice is recorded as `alba`; this spelling is preserved from the evaluation record.

[b] The Chinese and English voices are recorded as `zf_xiaobei` and `af_heart`, respectively.

[c] The Chinese and English voices are recorded as `hojo_zh_f_01` and `hojo_en_m_02`, respectively.

[d] The Chinese and English voices are recorded as `Xiaoya` and `Amy large`, respectively.

Seed-TTS evaluation. Table 2 shows that M5-TTS (GRPO) achieves 1.37% CER on Seed-ZH and 2.91% WER on Seed-EN, compared with 3.62% and 4.50% for MOSS-TTS-Nano. These correspond to relative error reductions of 62.2% and 35.3%, respectively, with half the language-model parameters. SIM is also higher on these subsets, at 0.65/0.53 versus 0.64/0.50. MOSS-TTS-Nano retains an advantage on Seed-Hard, with lower CER (15.77% vs. 17.48%) and higher SIM (0.63 vs. 0.59). Compared with the 170M-parameter Audio8-TTS Preview baseline, M5-TTS achieves lower Seed-Hard CER (17.48% vs. 19.63%), but has higher error rates on Seed-ZH and Seed-EN and lower SIM on all three subsets.

Multilingual voice cloning. On MiniMax (Table 3), M5-TTS (GRPO) achieves lower content error rates than MOSS-TTS-Nano in five of six languages. The relative reductions are 42.7%, 65.9%, 56.3%, 75.9%, and 64.9% for English, German, French, Japanese, and Korean, respectively. The largest reduction is observed in Japanese, where CER decreases from 40.22% to 9.70%. Chinese remains a gap, with 2.05% CER versus 1.80% for MOSS-TTS-Nano. SIM is higher in German, French, and Japanese, but lower in Chinese, English, and Korean. Pocket TTS achieves lower error rates in its reported English, German, and French voice-cloning evaluations, while M5-TTS has higher SIM in English and German and matches French SIM at the reported precision.

Table 3: Multilingual results on MiniMax. CER and WER are reported in percent. Lower CER/WER and higher SIM indicate better performance.

Model	Setting	ZH		EN		DE		FR		JA		KO	
		CER↓	SIM↑	WER↓	SIM↑	WER↓	SIM↑	WER↓	SIM↑	CER↓	SIM↑	CER↓	SIM↑
M5-TTS (CPT)	Voice Cloning	2.54	0.70	3.67	0.62	4.60	**0.63**	8.15	0.51	12.71	**0.72**	3.10	0.66
M5-TTS (GRPO)	Voice Cloning	2.05	0.71	3.06	0.63	3.81	**0.63**	8.14	**0.53**	9.70	**0.72**	2.88	0.66
M5-TTS	Custom Voice	1.16	—	2.24	—	**0.70**	—	5.33	—	10.27	—	**2.09**	—
MOSS-TTS-Nano	Voice Cloning	1.80	**0.73**	5.34	**0.65**	11.18	0.54	18.64	0.51	40.22	0.70	8.21	**0.67**
Pocket TTS	Voice Cloning	—	—	2.41	0.61	3.50	0.54	6.71	**0.53**	—	—	—	—
Pocket TTS	Custom Voice[a]	—	—	1.99	—	4.14	—	7.88	—	—	—	—	—
Sano TTS	Custom Voice[b]	1.41	—	**1.85**	—	3.18	—	8.91	—	—	—	—	—
Kokoro-TTS	Custom Voice[c]	**1.03**	—	2.58	—	—	—	**4.79**	—	**3.72**	—	—	—

Notes. — denotes an unavailable result or an unspecified setting. ZH, JA, and KO use CER; EN, DE, and FR use WER. All values are rounded to two decimal places. Setting distinguishes Voice-Cloning and Custom-Voice configurations. For models with multiple preset voices, Custom-Voice results are obtained using one selected voice per language, as specified below. Unmarked Custom-Voice models use a single fixed voice.
[a] The English, German, and French voices are recorded as `alba`, `juergen`, and `estelle`, respectively.
[b] The Chinese, English, German, and French voices are recorded as `Xiaoya`, `Amy large`, `Thorsten`, and `Siwis`, respectively.
[c] The Chinese, English, French, and Japanese voices are recorded as `zf_xiaobei`, `af_heart`, `ff_siwis`, and `jf_alpha`, respectively.

Effect of GRPO post-training. Relative to CPT, GRPO reduces error rates on all three Seed subsets: CER/WER decreases from 1.70% to 1.37% on Seed-ZH, from 3.60% to 2.91% on Seed-EN, and from 20.92% to 17.48% on Seed-Hard. The corresponding relative reductions are 19.4%, 19.2%, and 16.4%. On MiniMax, error rates decrease in all six languages, with the largest relative reduction in Japanese (23.7%); French changes only marginally, from 8.15% to 8.14% WER. Across both benchmarks, SIM is maintained or improved at the displayed precision. These comparisons evaluate the complete post-training recipe; they do not isolate the effects of the pause reward or local CER process feedback.

Speaker-specific synthesis. The custom-voice variant achieves 1.26% CER on Seed-ZH and 1.88% WER on Seed-EN. Compared with the GRPO voice-cloning variant, it also has lower content error rates in five of six MiniMax languages, including 0.70% WER in German. The GRPO voice-cloning variant performs better on Seed-Hard and MiniMax Japanese. Custom-voice results reflect dedicated speaker configurations and are reported separately from zero-shot voice cloning.

Dialect synthesis. Table 4 evaluates Cantonese and Sichuanese, using the same ASR evaluator for all systems within each dialect. For Cantonese, M5-TTS achieves the lowest CER among the compared systems on both subsets, at 8.55% on Easy and 8.70% on Hard. Relative to CosyVoice2-Yue, these correspond to reductions of 17.2% and 8.3%; relative to Llasa-1B-Yue, the reductions are 21.5% and 31.9%. Its SIM is lower than both baselines on both subsets, indicating a remaining gap in speaker preservation. For Sichuanese, M5-TTS reduces Hard CER from 8.78% to 7.78% relative to CosyVoice2-WSC, an 11.4% relative reduction, but has higher Easy CER (5.08% vs. 4.28%) and lower SIM on both subsets. These results demonstrate strengths in content accuracy on Cantonese and the Sichuanese Hard subset, alongside remaining limitations in speaker similarity.

4.3 Subjective Evaluation

We compare M5-TTS with MOSS-TTS-Nano and Kokoro-TTS using four mean opinion score (MOS) metrics: N-MOS for speech naturalness, S-MOS for voice-cloning similarity, I-MOS for speech intelligibility, and P-MOS for prosody and expressiveness. N-MOS measures how natural and human-like the synthesized speech sounds; S-MOS evaluates perceived resemblance to the reference speaker; I-MOS assesses how easily the speech can be understood; and P-MOS evaluates prosodic delivery and expressiveness. The evaluation involves 10 listeners, who assign scores from 1 to 5, with higher scores indicating better performance. We report the mean ratings for each dimension.

As shown in Table 5, M5-TTS receives higher speaker-similarity and intelligibility ratings than MOSS-TTS-Nano, with S-MOS increasing from 2.85 to 3.05 and I-MOS from 2.65 to 2.95. Its P-MOS is also slightly higher, at 2.85 versus 2.80. These ratings suggest benefits in perceived speaker resemblance and intelligibility, alongside a modest numerical improvement in prosody and expressiveness. However, M5-TTS receives a lower N-MOS than MOSS-TTS-Nano (2.90 vs. 3.00), indicating that perceived naturalness

Table 4: Results on the Easy and Hard subsets for Cantonese (Yue) and Sichuanese (Chuan).

Dialect	TTS Model	ASR Model	Easy		Hard	
			CER↓	SIM↑	CER↓	SIM↑
Yue	M5-TTS (GRPO)	U2pp-Conformer-Yue	**8.55**	0.47	**8.70**	0.51
	CosyVoice2-Yue	U2pp-Conformer-Yue	10.33	**0.62**	9.49	**0.67**
	Llasa-1B-Yue	U2pp-Conformer-Yue	10.89	0.51	12.78	0.55
Chuan	M5-TTS (GRPO)	FireRedASR-AED	5.08	0.58	**7.78**	0.59
	CosyVoice2-WSC	FireRedASR-AED	**4.28**	**0.64**	8.78	**0.65**

Table 5: Subjective evaluation involving 10 listeners. All metrics use a five-point scale; higher scores are better. The highest reported score in each column is shown in bold.

Model	N-MOS↑	S-MOS↑	I-MOS↑	P-MOS↑
Kokoro-TTS	2.85	—	**3.10**	2.60
MOSS-TTS-Nano	**3.00**	2.85	2.65	2.80
M5-TTS	2.90	**3.05**	2.95	**2.85**

Notes. N-MOS: naturalness; S-MOS: speaker similarity; I-MOS: intelligibility; P-MOS: prosody and expressiveness. — indicates an unreported score.

remains an area for improvement.

Compared with Kokoro-TTS, M5-TTS obtains a slightly higher N-MOS (2.90 vs. 2.85) and a higher P-MOS (2.85 vs. 2.60), suggesting more expressive prosodic delivery in the evaluated samples. Kokoro-TTS achieves the highest I-MOS, at 3.10, exceeding the 2.95 obtained by M5-TTS. M5-TTS has the highest reported S-MOS and P-MOS, MOSS-TTS-Nano leads in naturalness, and Kokoro-TTS leads in intelligibility. These are descriptive differences; without uncertainty estimates, the small score gaps do not establish statistical significance.

4.4 Efficiency Evaluation

We evaluate the inference efficiency of M5-TTS using two key metrics: first-chunk latency and real-time factor (RTF). First-chunk latency measures the elapsed time from the initiation of speech generation to the availability of the first audio chunk, reflecting the system's responsiveness in streaming synthesis. RTF is defined as the ratio of total synthesis time to the duration of the generated speech, measuring the overall computational efficiency of speech generation. An RTF below 1 indicates that speech can be synthesized faster than its playback duration, a key requirement for real-time applications.

On four cores of an Intel Xeon Gold 6530, M5-TTS achieves an RTF of 0.653 and a warm first-chunk latency of 225 ms. This corresponds to approximately 1.53 seconds of generated speech per second of computation. The timing excludes model loading and warmup and demonstrates faster-than-real-time streaming in the evaluated server-CPU configuration; it does not establish performance on mobile or ARM devices.

5 Limitations and Future Work

The current 16-codebook speech representation introduces considerable inference overhead in the local autoregressive decoder. Repeated codebook-generation steps require frequent model-weight and cache accesses and can place substantial demands on memory bandwidth, limiting CPU inference efficiency despite the compact LM. We plan to investigate inference with eight codebooks to reduce local decoding overhead, while evaluating the resulting trade-off between computational efficiency and acoustic fidelity.

Audio quality also remains a limitation. The current system supports only 24 kHz output, and highly expressive synthesis can exhibit voice breaks and hoarse or muted segments. These artifacts reduce perceived audio quality and indicate that expressive acoustic generation requires further refinement. Future work will extend the system to support 48 kHz output and improve the robustness of acoustic generation under expressive speaking conditions.

Intelligibility remains challenging on difficult inputs. M5-TTS (GRPO) achieves a CER of 17.48% on Seed-Hard, compared with 15.77% for MOSS-TTS-Nano, indicating that its pronunciation robustness

under challenging conditions is still insufficient. Addressing this gap requires continued refinement of the data pipeline and training-data mixture. We will improve data quality control, expand coverage of challenging pronunciation patterns, and refine sampling proportions across general and difficult examples to strengthen robustness while preserving performance on standard multilingual inputs.

We also plan to introduce explicit control over emotion, paralinguistic expressions, and speaking style, including controllable emotional delivery and nonverbal vocalizations such as laughter and sighs. These extensions aim to enable more expressive and context-appropriate speech generation for on-device applications and embodied voice interaction.

6 Conclusion

We presented M5-TTS, a compact speech generation framework built around a 50M-parameter language model, excluding the speech codec. Its hierarchical architecture combines a 16-codebook speech representation with multi-token prediction. The system supports multilingual and multidialect synthesis, reference-based zero-shot voice cloning, and speaker-specific multilingual generation. GRPO post-training adds a token-level CER process advantage to global speaker-similarity and pause feedback. Character-level error comparisons, backward negative-credit propagation, and temporal projection provide local content feedback to the codec-token policy.

Experiments demonstrate strong content accuracy within this compact parameter budget. Using half the LM parameters of MOSS-TTS-Nano, M5-TTS achieves relative error reductions of 62.2% on Seed-ZH and 35.3% on Seed-EN, and lower content error rates in five of the six MiniMax languages. Compared with CPT, GRPO reduces content error rates across the reported Seed and MiniMax subsets while maintaining or improving speaker similarity at the reported precision. Dialect evaluations further show lower Cantonese CER than both compared baselines and improved performance on the Sichuanese Hard subset, although challenging Chinese utterances and dialectal speaker preservation remain areas for improvement.

On four cores of an Intel Xeon Gold 6530, M5-TTS achieves an RTF of 0.653 and a warm first-chunk latency of 225 ms, demonstrating faster-than-real-time synthesis with responsive streaming output in the evaluated CPU configuration. These results support M5-TTS as a compact speech generation component for interactive applications and motivate further deployment and evaluation on resource-constrained edge devices for embodied voice interaction.

7 Ethics Statement

Voice cloning and speaker-specific synthesis can support accessibility, language learning, and personalized interfaces, but can also facilitate impersonation, fraud, harassment, or misleading recordings. Use of M5-TTS should require appropriate authorization from the represented speaker and respect applicable identity, publicity, and intellectual-property rights. Synthetic speech should be disclosed where listeners could otherwise reasonably mistake it for an authentic recording. The system should not be used to attribute statements to a person deceptively or to evade identity verification.

Training data and reference recordings raise separate licensing and privacy concerns. Data use and redistribution should follow the permissions and restrictions of each source, including those governing derived checkpoints. Reference audio, transcripts, and speaker representations should be handled as potentially identifying information, with collection, retention, and sharing limited to their stated purpose. Local inference can reduce transmission to external services, but does not by itself guarantee privacy.

Coverage and synthesis quality may differ across languages, dialects, accents, and speaker groups. The reported benchmark results should not be treated as proof of equal performance or suitability for consequential applications.

A Implementation and Reproducibility Details

A.1 Conditioning and Codebook Dependencies

In the speaker-specific setting, the conditioning prefix orders accent, speaker, and language identifiers before the text. Pre-training omits the speaker field. The following listing preserves the special-token spellings; line breaks are removed when concatenating the prefix. Braced fields denote values to substitute, rather than literal text to synthesize.

```
<|accent_begin|>{accent id}<|accent_end|>
<|speaker_start|>{speaker id}<|speaker_end|>
<|lang_begin|>{lang id}<|lang_end|>{text}
```

This prefix specifies the ordering of the three controls; it does not specify the full serialization of reference speech for voice cloning. Identifier inventories, text normalization, and reference-prompt construction should accompany the released inference implementation. No additional codec boundary tokens are implied by this notation.

Let $q_{t,c}$ denote codebook c at audio frame t, with $c \in \{0, \ldots, 15\}$. The temporal backbone consumes the conditioning sequence and completed earlier frames. Within a historical frame, the 16 codebook embeddings are summed into one frame-level input. Its hidden state h_t first predicts $q_{t,0}$; the local MTP Transformer subsequently generates $q_{t,1}, \ldots, q_{t,15}$ sequentially:

$$p(\mathbf{q}_t \mid h_t) = p(q_{t,0} \mid h_t) \prod_{c=1}^{15} p(q_{t,c} \mid h_t, q_{t,0:c-1}). \tag{7}$$

Thus, the residual codebooks are not 15 independent parallel predictions. The backbone cache persists across frames, whereas the local cache is reset for each new frame. The local Transformer predicts codec indices; the separate speech-codec decoder reconstructs the waveform. The 50M parameter count refers to the LM component and excludes that codec.

A.2 Pre-training and Continued Pre-training Hyperparameters

Table 6 summarizes the configurations used for pre-training and continued pre-training (CPT). Both stages share the same backbone, local decoder, tokenizer, and 16-codebook speech representation, and use BF16 precision on CUDA. CPT uses lower learning rates, a shorter learning-rate schedule, and a larger maximum token budget. The `tokenizer` field refers to text tokenization; the speech tokenizer is described in Section 2.

Table 6: Hyperparameter configurations for pre-training and CPT. Field names follow the training configuration.

Hyperparameter	Pre-training	CPT
`backbone_flavor`	`llama-30M`	`llama-30M`
`decoder_flavor`	`llama-20M`	`llama-20M`
`tokenizer`	`llama3`	`llama3`
`audio_num_codebooks`	16	16
`learning_rate`	1×10^{-4}	1×10^{-5}
`min_lr`	1×10^{-5}	1×10^{-6}
`scheduler_steps`	1,000,000	150,000
`num_warmup_steps`	4,000	1,000
`save_every`	500,000	15,000
`max_tokens`	13,000	25,000
`max_batch_size`	999	999
`pad_multiple`	16	16
`decoder_fraction`	0.125	0.125
`lambda_depth`	1.0	1.0
`device`	`cuda`	`cuda`
`precision`	`bf16`	`bf16`

A.3 GRPO Reward and Loss Details

Pause score. For adjacent aligned characters, let $s_{i,j}$ and $u_{i,j}$ denote the start and end timestamps of character j in candidate i. The gap is $d_{i,j} = \max(0, s_{i,j+1} - u_{i,j})$. Ordinary boundaries and boundaries associated with punctuation use different duration constraints:

$$r_{i,j}^{\text{pause}} = \begin{cases} \mathbb{I}(d_{i,j} \le \tau_{\text{intra}}), & j \in \mathcal{B}_{i,\text{intra}}, \\ \mathbb{I}(\tau_{\text{punct}}^{-} \le d_{i,j} \le \tau_{\text{punct}}^{+}), & j \in \mathcal{B}_{i,\text{punct}}. \end{cases} \tag{8}$$

The utterance score averages these indicators over the valid aligned boundaries $\mathcal{B}_i$:

$$R_i^{\text{pause}} = \frac{1}{|\mathcal{B}_i|} \sum_{j \in \mathcal{B}_i} r_{i,j}^{\text{pause}}. \tag{9}$$

The thresholds define the allowed intra-sentence gap and the acceptable punctuation-pause range. This score is an utterance-level reward; local content feedback is constructed separately from character edit counts.

Illustrative credit propagation. Consider four characters in one clause with unpropagated advantages $(a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}) = (0, 0, -1, 0)$. Equation 3 gives $\widetilde{\mathbf{a}}_i = (-e^{-2}, -e^{-1}, -1, 0)$. For frames fully contained in the respective character intervals, the local contributions added to the global advantage are approximately $(-0.068, -0.184, -0.5, 0)$ at $\lambda_{\text{CER}} = 0.5$. The first two characters receive attenuated feedback from the third, although their original local advantages were zero. The fourth character receives no local contribution in this example. This illustrates credit propagation; the values are not measured training results.

Multi-codebook loss aggregation. The first-codebook policy loss is normalized by the number of generated decisions. Local-decoder losses are weighted across codebooks on sampled frames and normalized by the sampled-frame count. The two components are added, and the same aggregation is applied to the token-wise KL estimates. Valid reference-prefix positions have zero policy advantage and contribute only to KL regularization. No length-based scaling or additional clipping is applied to the local CER advantage, as recorded in Table 7.

A.4 GRPO Hyperparameters

Table 7 summarizes the GRPO configuration. Each global batch contains 32 prompts, with eight candidates generated per prompt, yielding 256 candidate utterances. All eight candidates are retained without rollout selection. Utterance-level SIM and pause rewards have weights of 1.0 and 0.5, respectively. Utterance-level CER broadcasting is disabled; the projected CER process advantage is added separately with a weight of 0.5, following Equation 4. The KL regularization coefficient is 0.01.

Table 7: Hyperparameters for GRPO post-training. Field names follow the training configuration.

Hyperparameter	Value
Rewards and optimization	
reward_weight.cer_reward	0.0
reward_weight.sim_reward	1.0
reward_weight.pause_reward	0.5
token_cer.enabled	true
token_cer.mode	char_exp
token_cer.weight	0.5
token_cer.length_scale	false
token_cer.advantage_clip	null
token_cer.ambiguous	use_path
use_gdpo	false
kl_weight	0.01
Batching and rollout	
batch_size	32
batch_size_scope	global
batch_size_per_rank	16
num_answers_per_question	8
rollout_micro_batch_size	8
rollout_selection.enabled	false
rollout_selection.keep_per_group	8
rollout_mode	remote
async_rollout	true
max_policy_lag	1
resample_max_attempts	3
Generation settings	
generation_conf.temperature	1.0
generation_conf.topk	50
generation_conf.max_audio_length_ms	40,000
generation_conf.max_seq_len	2,048
Reward execution and reproducibility	
async_reward	true
reward_workers	8
reward_max_inflight	16
reward_timeout	300
seed	10,086

Notes. Reward workers are HTTP threads per actor. Rollout selection is disabled, so its configured selection method is inactive.

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